MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

**Project** Mansourah Massarah Gold Project
**Customer:** MA’ADEN GOLD AND BASE METALS (MA’ADEN)

**Title:**

# GENERAL ARRANGEMENT
# POWER PLANT MISCELLANEOUS STRUCTURE
# G.A. AND DETAIL

**Document Number:** MD-216-49PR-EG-CV-GAD-2025
**Revision:** D01
**Revision Date:** 17.02.2023
**Status:** AS BUILT

| | | | | | | | |
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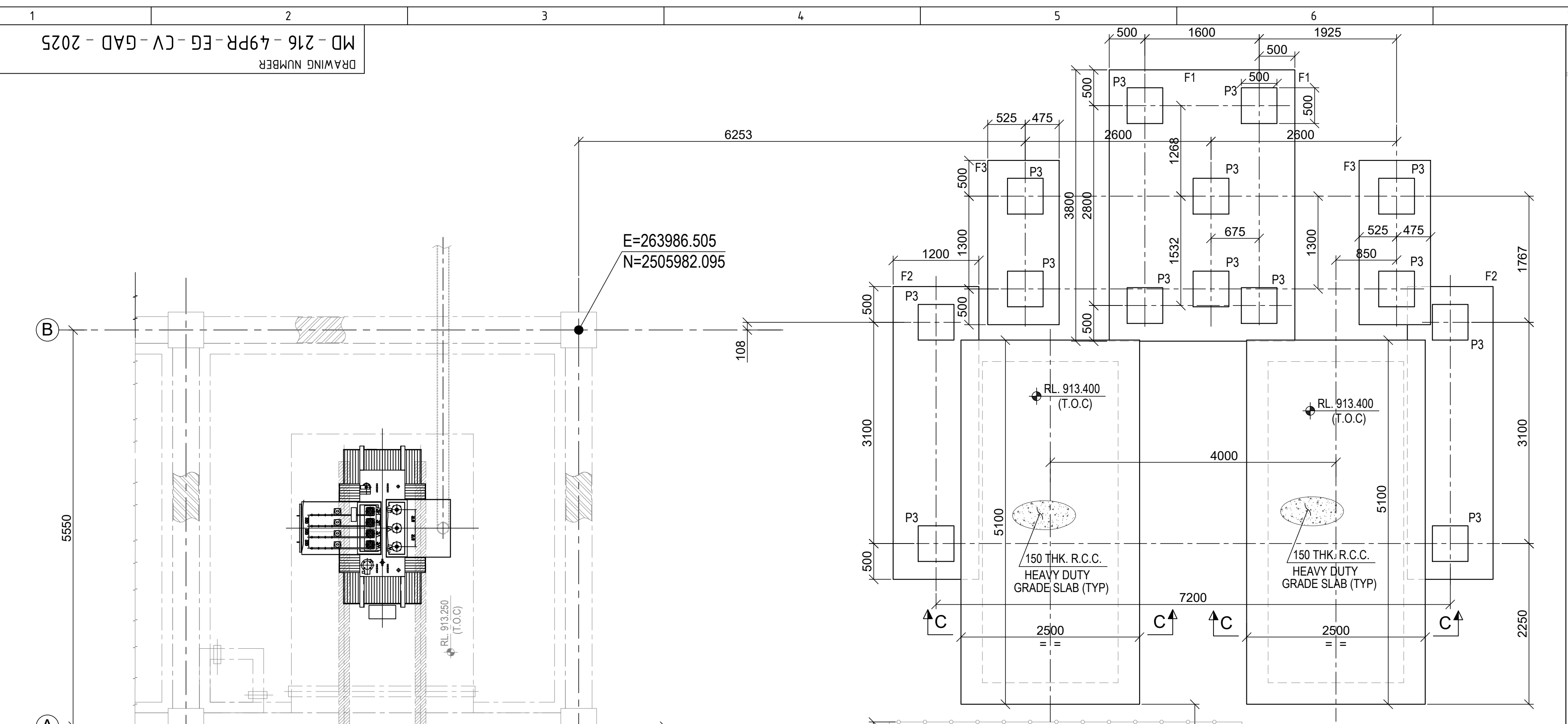
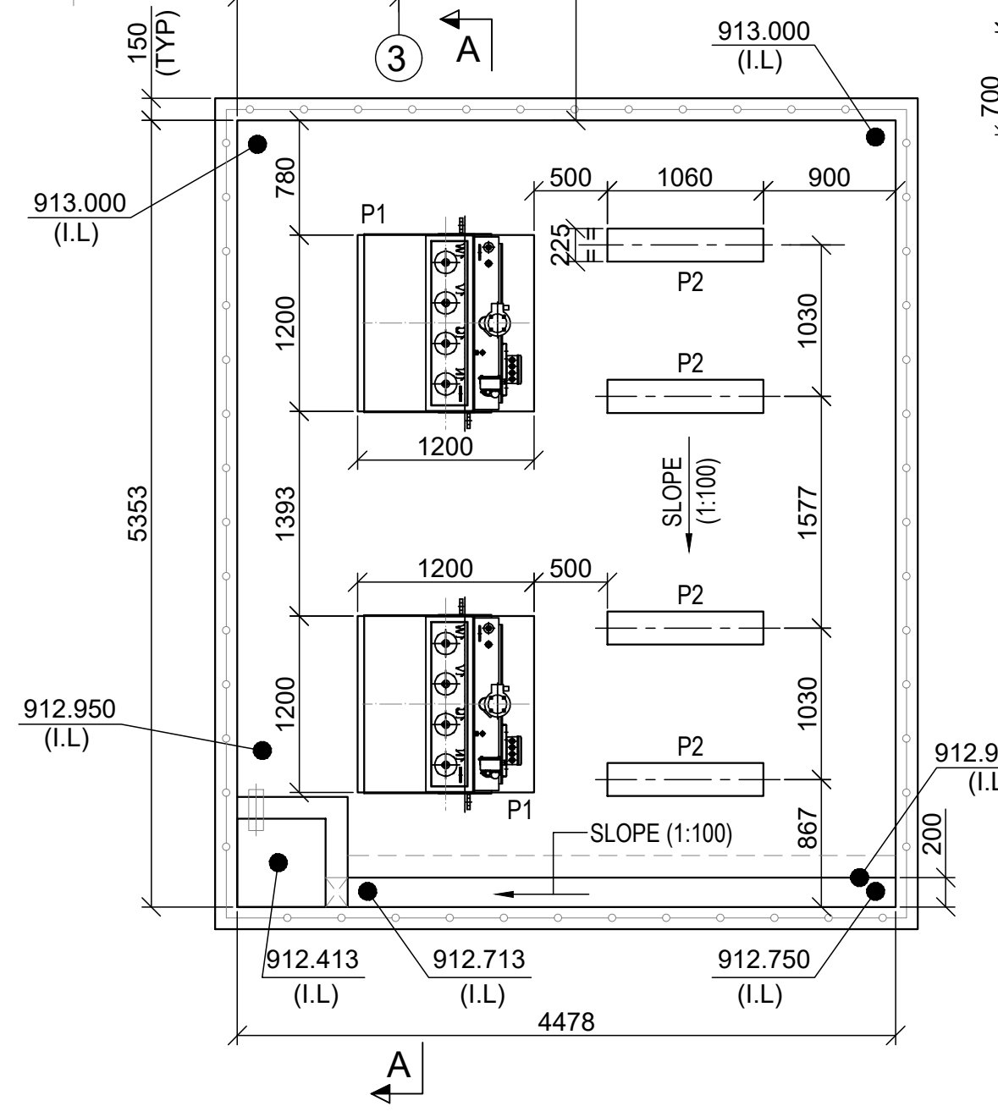
| L&T Document Number: | O19045 | 008 | C | G | 24298 | 007 | 6 |

**REVISION HISTORY**

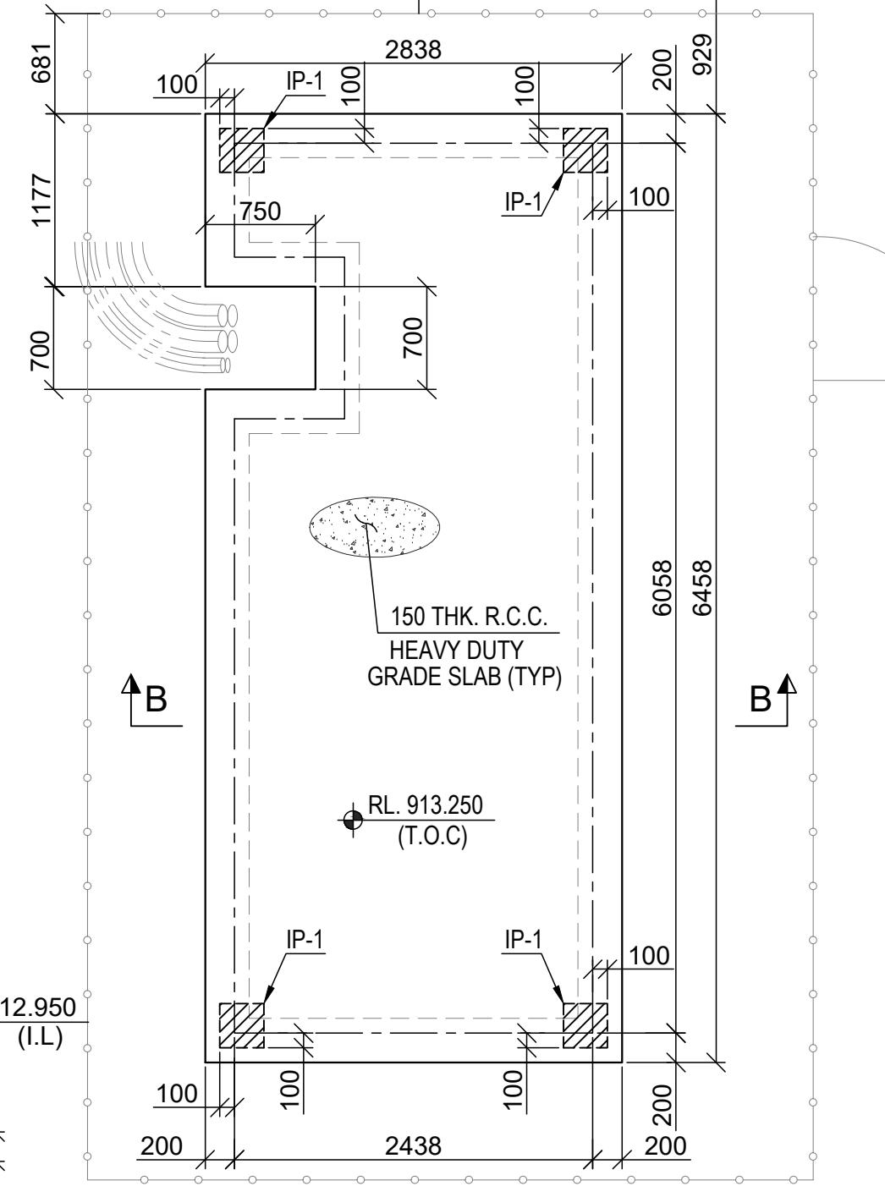
| Rev | Date | Revision Text | Prepared By | Reviewed By | Approved By | Employer |
|---|---|---|---|---|---|---|
| A01 | 30.11..2020 | ISSUED FOR REVIEW | PDK | SGC | PRI | |
| C01 | 04.01..2021 | ISSUED FOR CONSTRUCTION & REVISED AS MKD. △C01 | PDK | SGC | PRI | |
| C02 | 19.03.2021 | ISSUED FOR CONSTRUCTION & REVISED AS MKD. △C02 | PDK | SGC | PRI | |
| C03 | 26.03.2021 | ISSUED FOR CONSTRUCTION & REVISED AS MKD. △C03 | AMJ | SGC | PRI | |
| C04 | 03.07.2021 | ISSUED FOR CONSTRUCTION & REVISED AS MKD. △C04 | AMJ | SGC | PRI | |
| C05 | 21.07.2021 | ISSUED FOR CONSTRUCTION & REVISED AS MKD. △C05 | AMJ | SGC | PRI | |
| D01 | 17.02.2023 | AS BUILT | AMJ | SGC | PRI | |

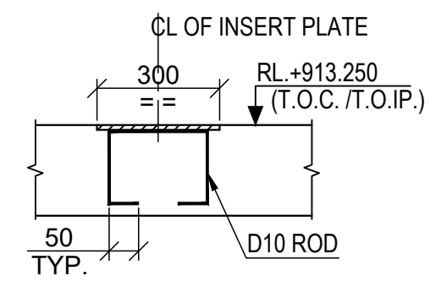




## PLAN OF TRANSFORMER FOUNDATION

(SCALE 1:50)





## SECTION : D - D

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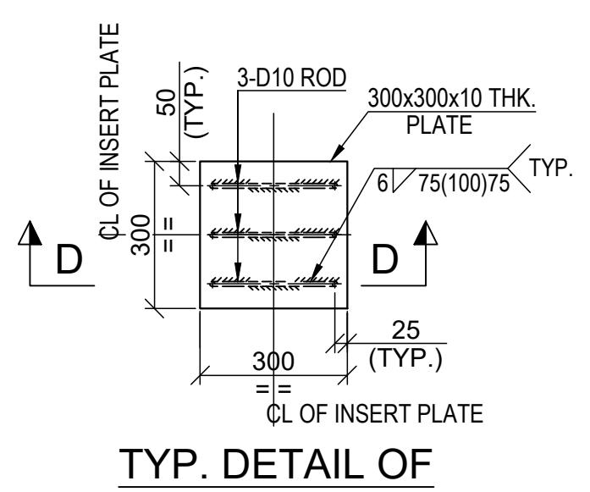
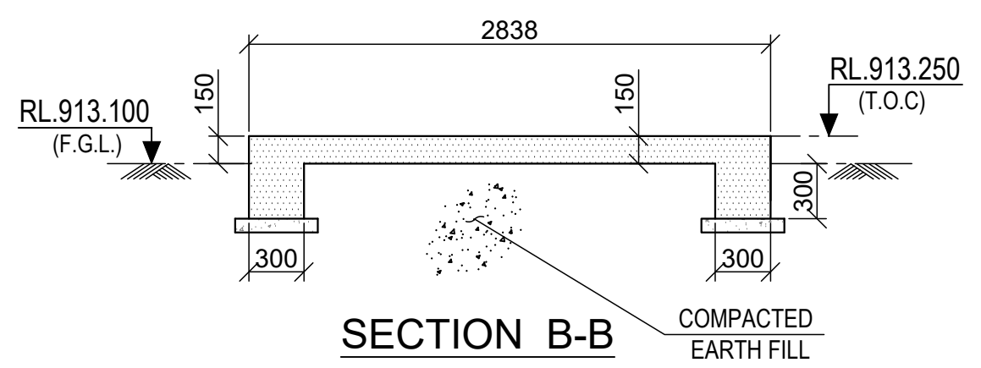


## TYP. DETAIL OF INSERT PLATE MKD. IP-1

(SCALE 1:15)



## SECTION B-B

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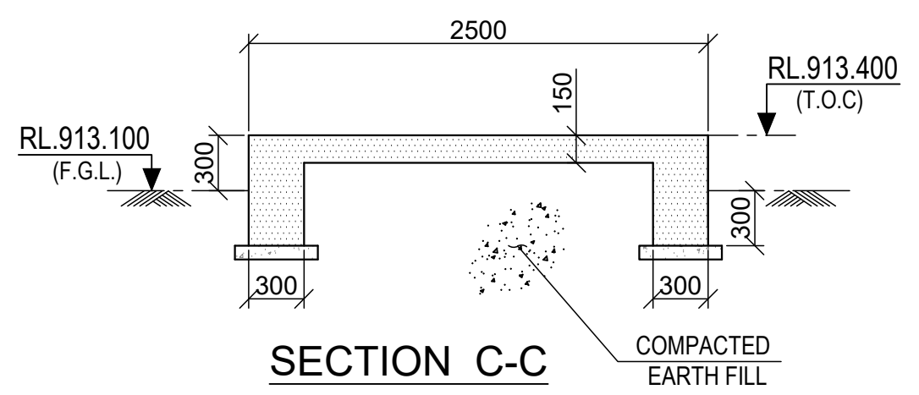

## SECTION C-C

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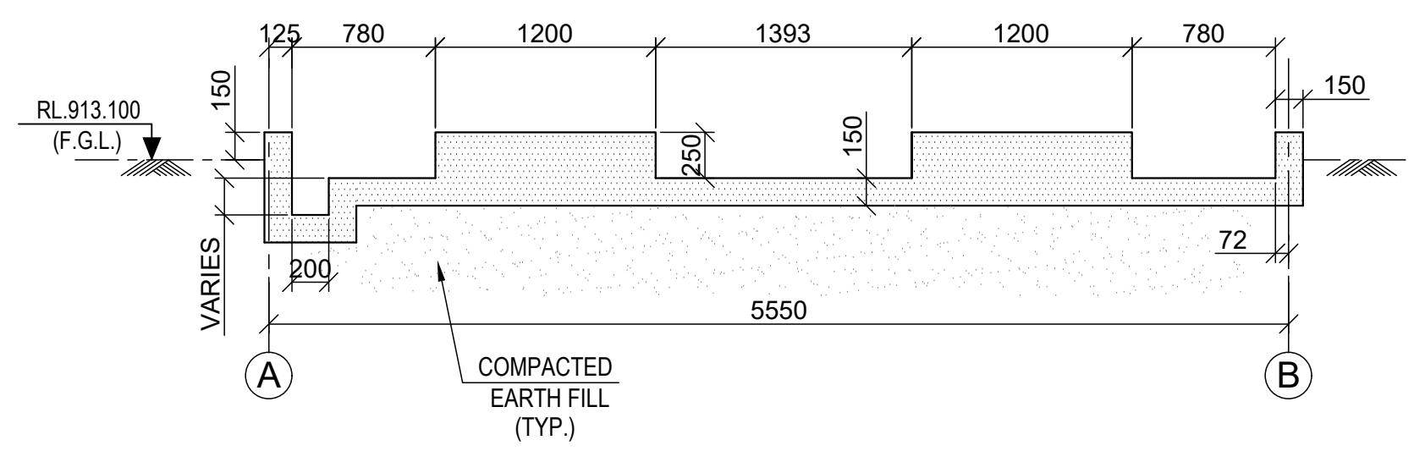
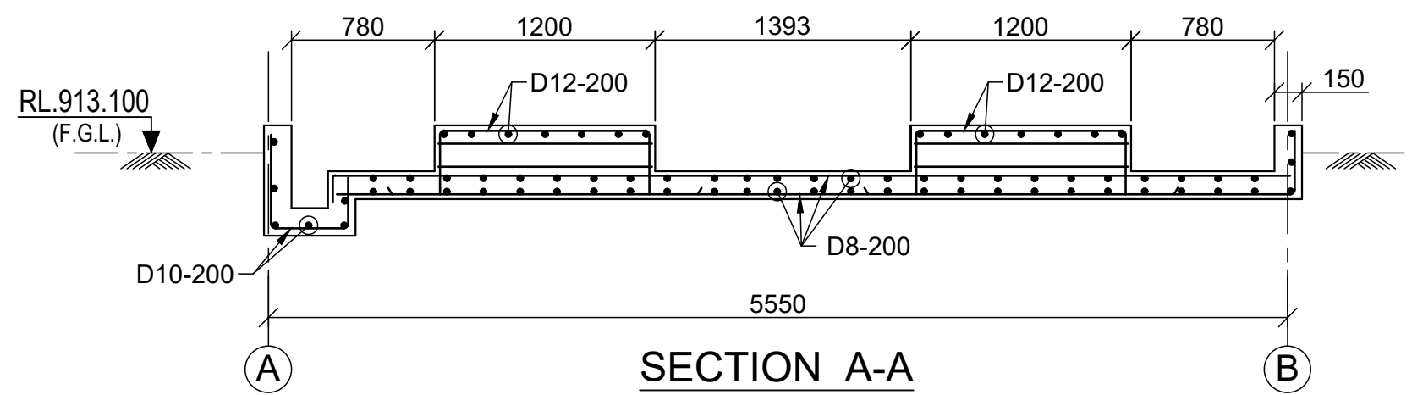


## SECTION A-A

(SCALE - 1:40)



## SECTION A-A

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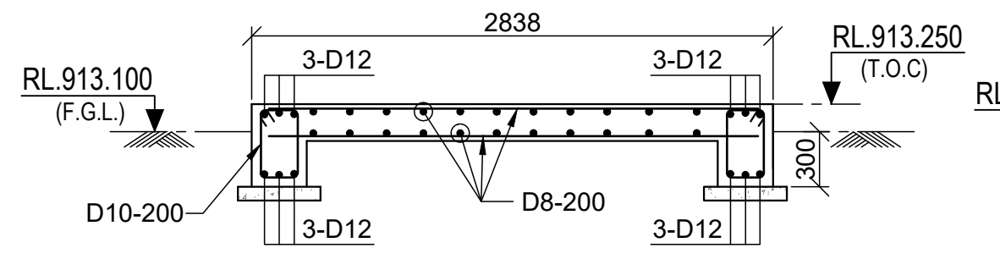
(SCALE - 1:40)



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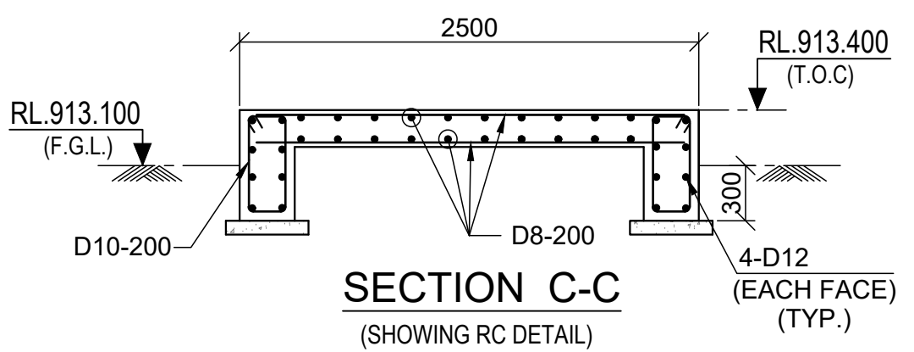

## SECTION C-C

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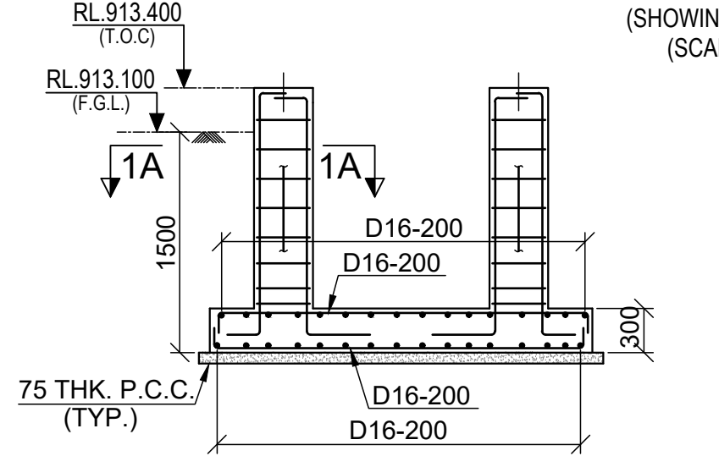

## SECTION 1-1

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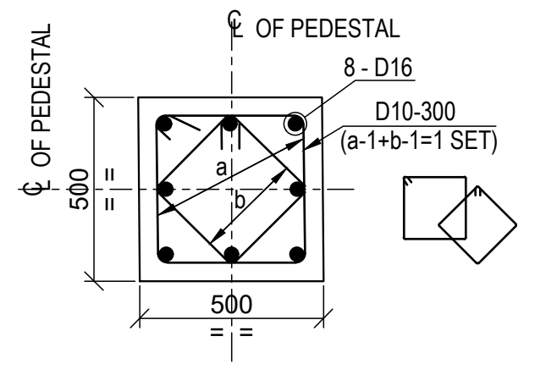

## SECTION : 1A - 1A

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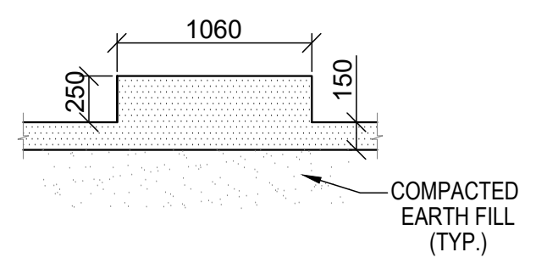

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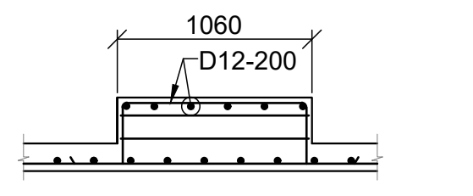
(SCALE - 1:40)



## DETAIL OF PEDESTAL MKD. P2

(SHOWING RC DETAIL)
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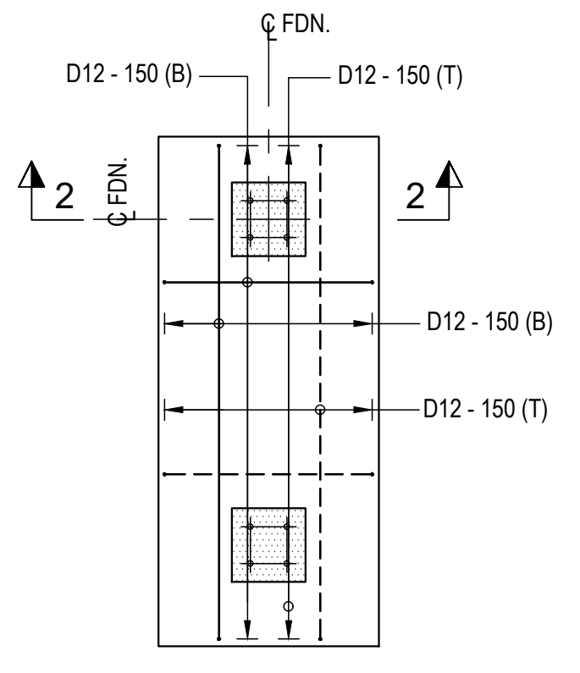
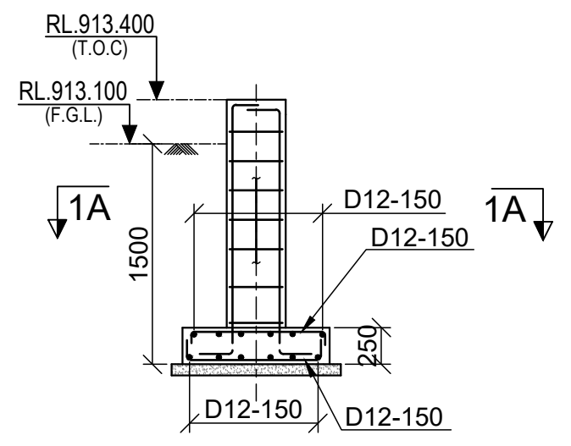


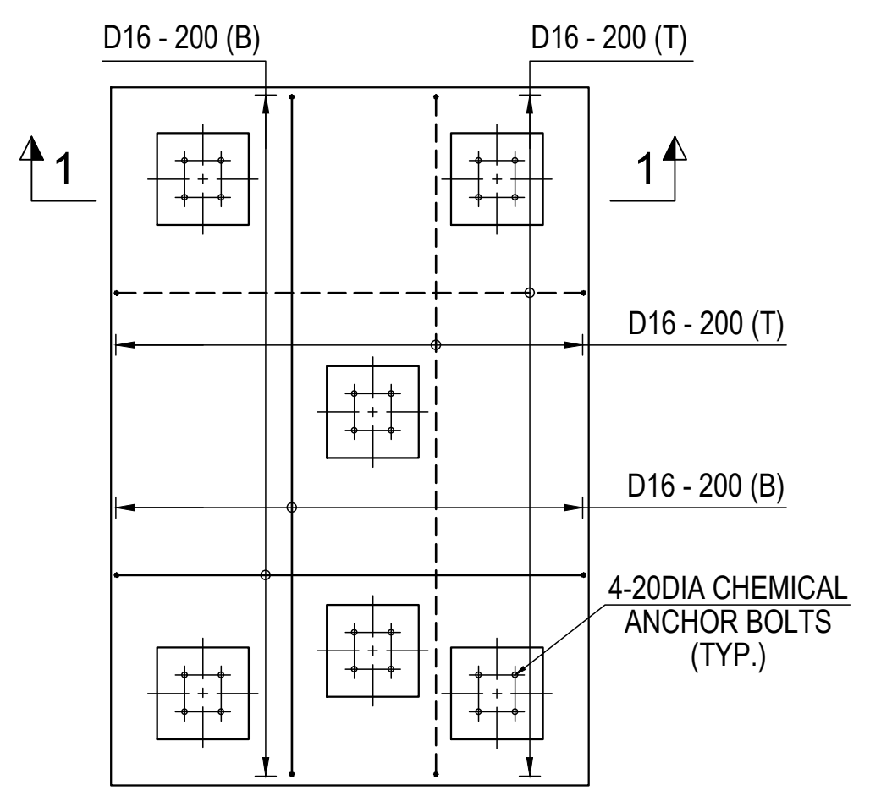
## DETAIL OF FOOTING MKD. F2

(SHOWING RC DETAIL)
(SCALE - 1:40)



## SECTION 2-2

( SCALE - 1:50 )



## DETAIL OF FOOTING MKD. F1

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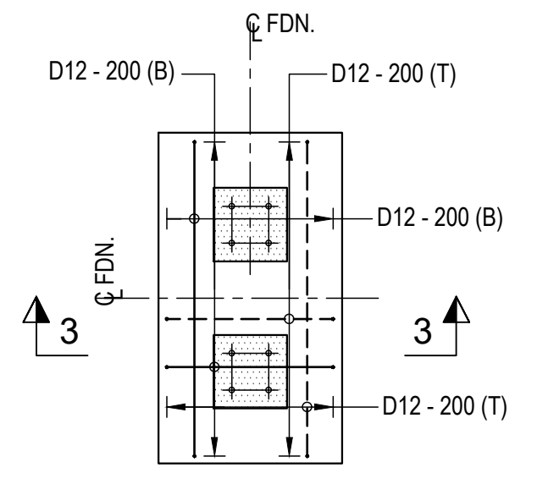

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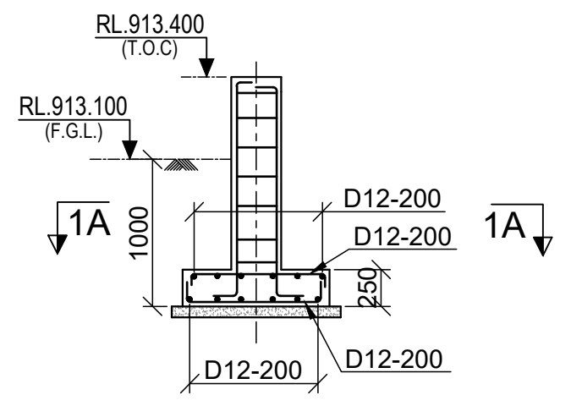
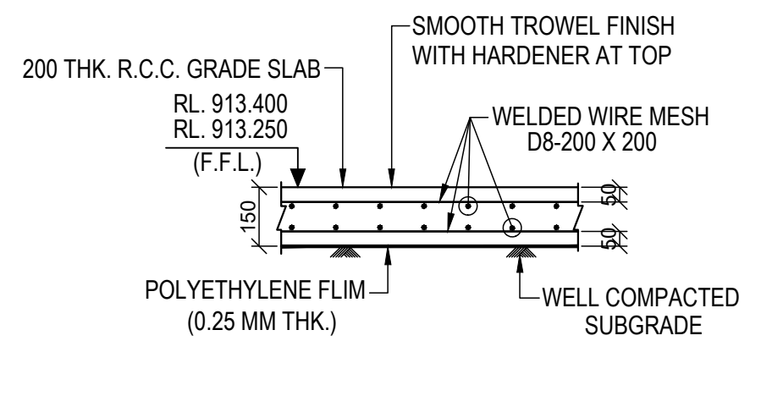
(SHOWING RC DETAIL)
(SCALE - 1:40)



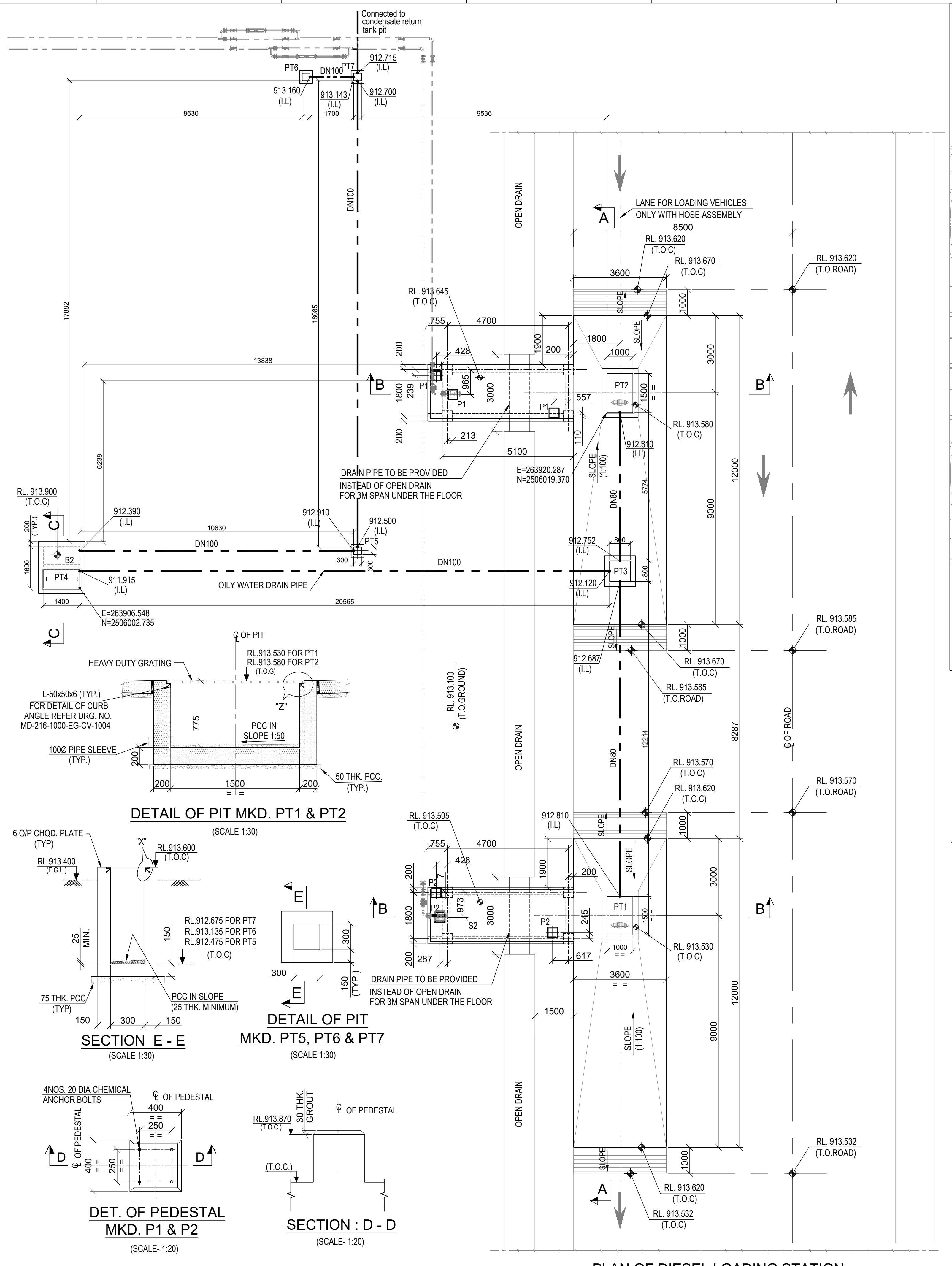
## SECTION 3-3

( SCALE - 1:50 )



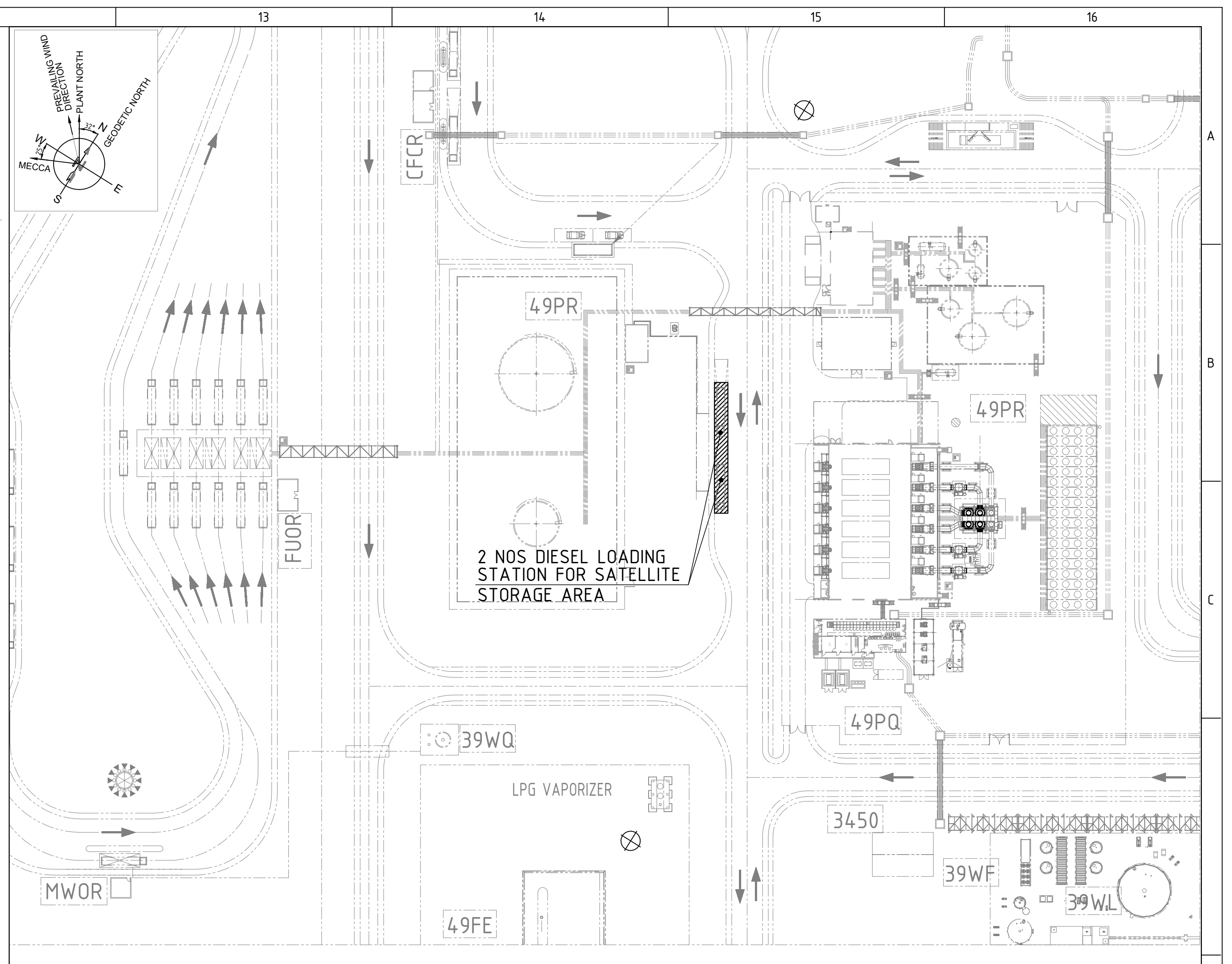
## TYP. RC DETAIL OF 150 THK. HEAVY DUTY GRADE SLAB

SCALE 1:25



## PLAN OF DIESEL LOADING STATION

(SCALE 1:100)



## KEY PLAN

(SCALE 1:1000)

## DETAIL OF PIT MKD. PT1 & PT2

(SCALE 1:30)

## SECTION E - E

(SCALE 1:30)

## DETAIL OF PIT MKD. PT5, PT6 & PT7

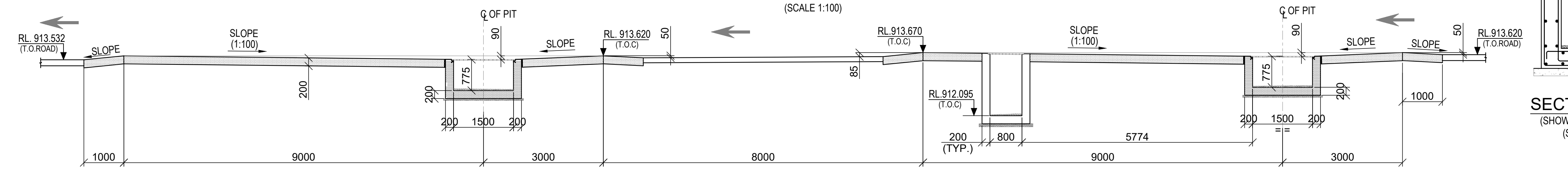
(SCALE 1:30)

## DET. OF PEDESTAL MKD. P1 & P2

(SCALE- 1:20)

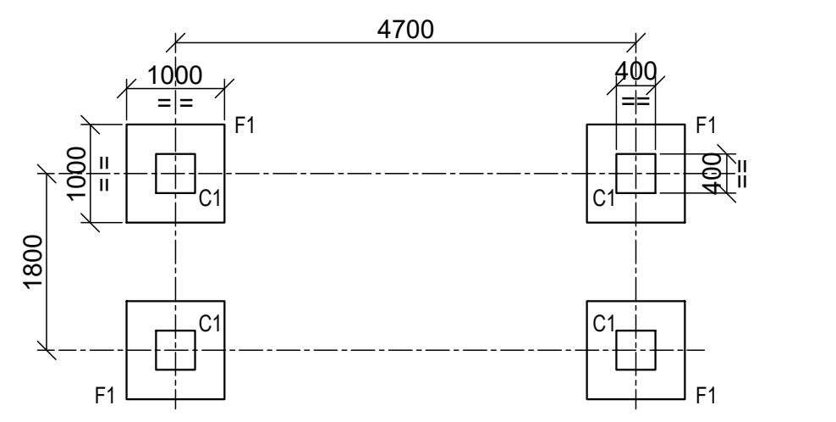
## SECTION : D - D

(SCALE- 1:20)



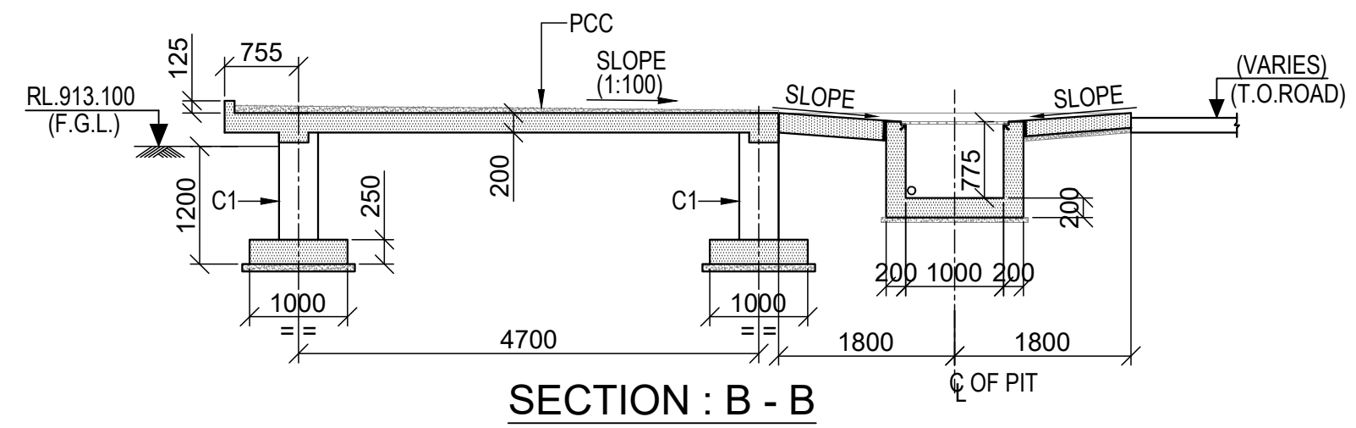
## SECTION : A - A

(SCALE 1:75)



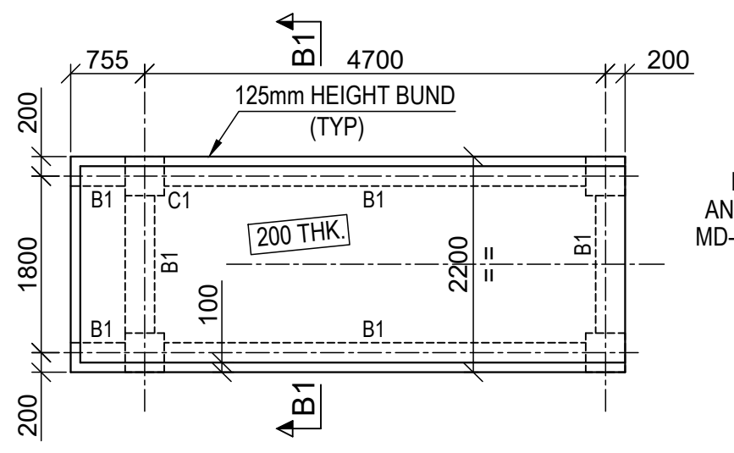
## FOUNDATION PLAN AT RL.911.900 (B.O.C.)

(SCALE 1:75)



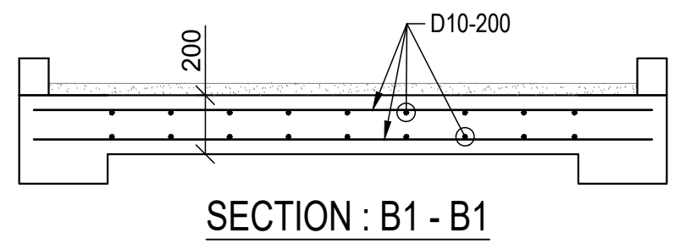
## SECTION : B - B

(SCALE 1:75)



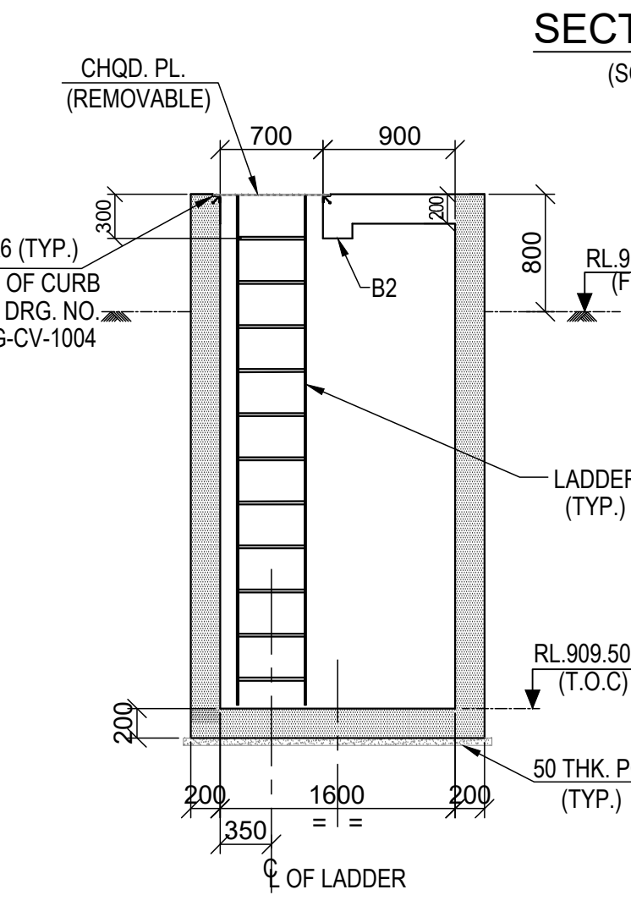
## PLAN AT SLAB LEVEL

(PEDESTAL NOT SHOWN HERE)
(SCALE 1:75)



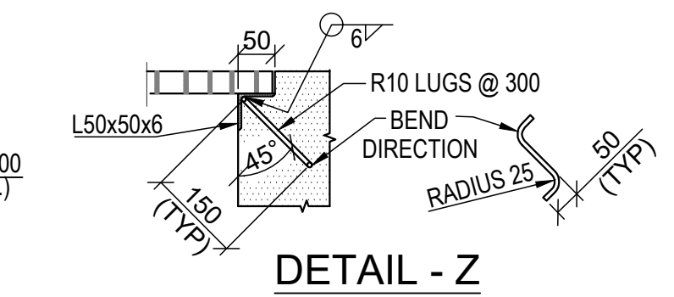
## SECTION : B1 - B1

(SHOWING REINFORCEMENT ONLY)
SCALE 1:25



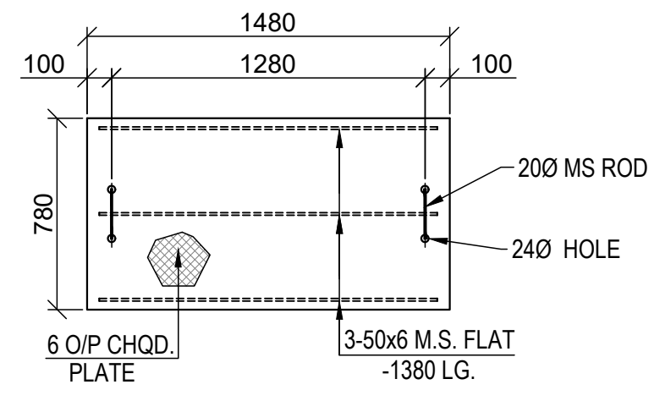
## SECTION C-C

(SCALE 1:50)



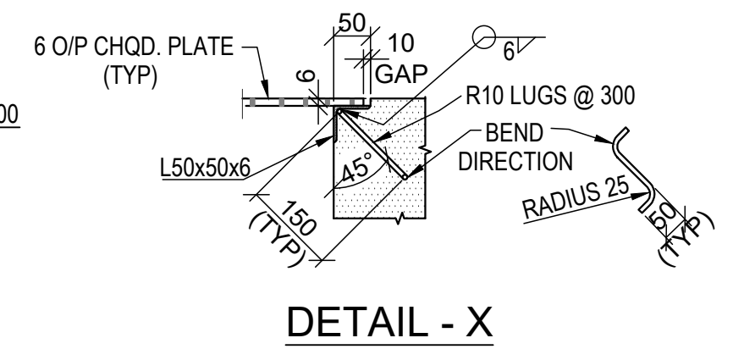
## DETAIL - Z

(SCALE 1:10)



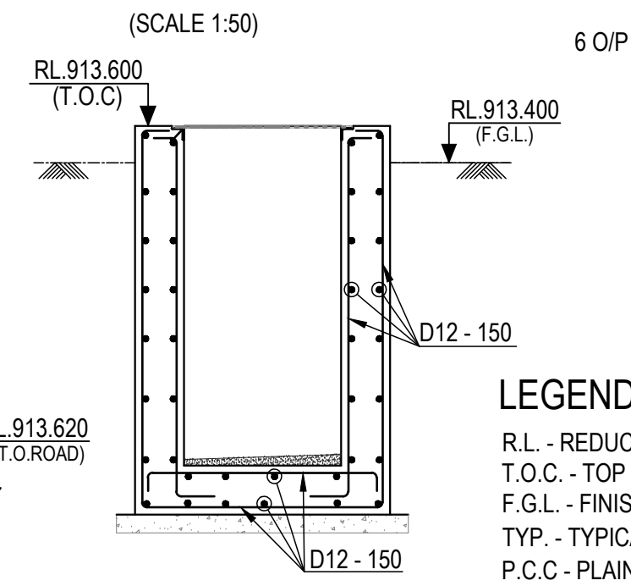
## DETAIL OF CHQD. PLATE

(SCALE- 1:30)



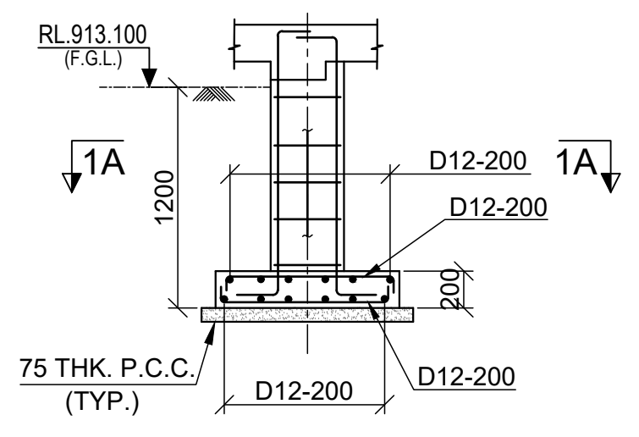
## DETAIL - X

(SCALE 1:10)



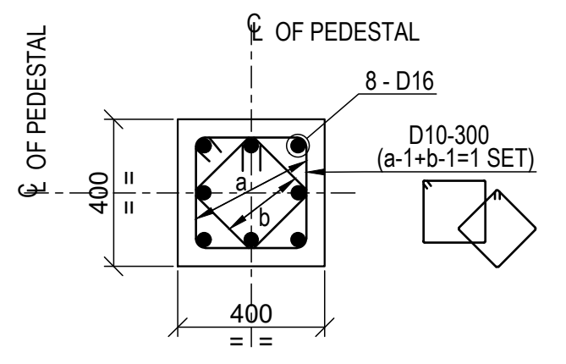
## SECTION E - E

(SHOWING RC DETAIL)
(SCALE 1:30)



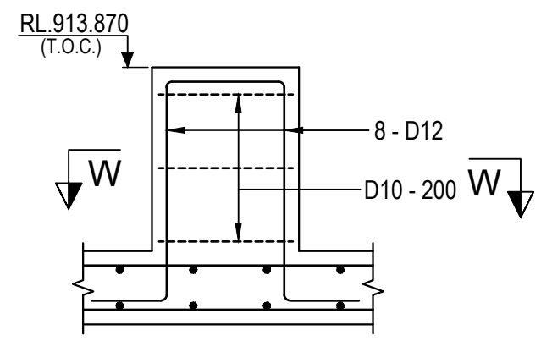
## SECTION 1-1

( SCALE - 1:50 )



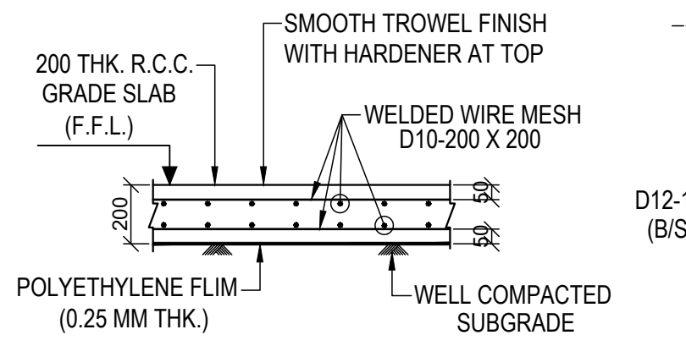
## SECTION : 1A - 1A

(TYP. RC DETAIL OF PEDESTAL MKD. C1)
(SCALE- 1:20)



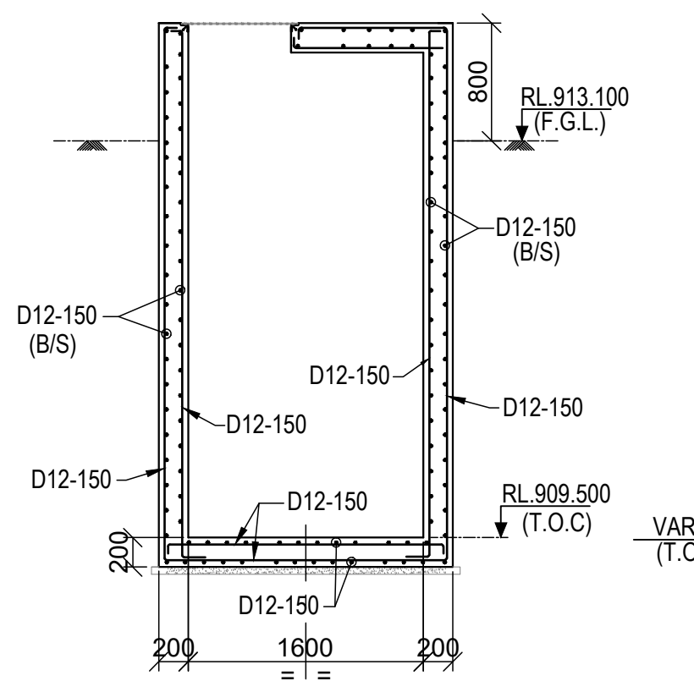
## R.C. DET. OF PEDESTAL MKD. P1 & P2

(SCALE- 1:20)



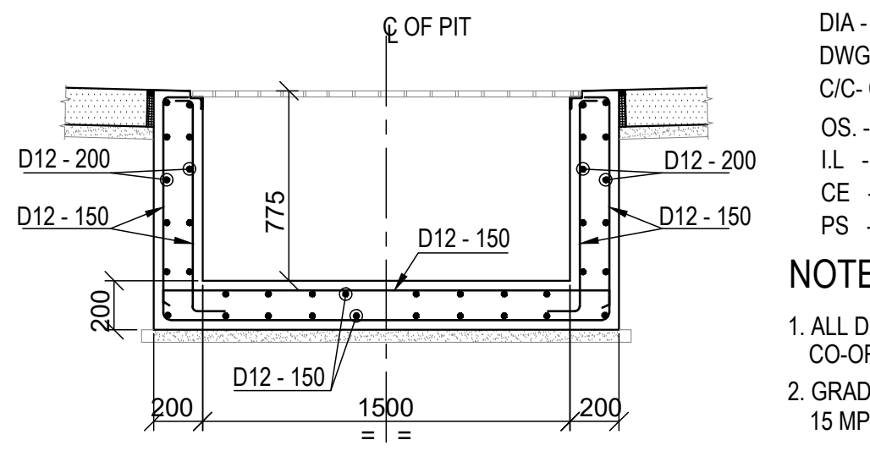
## TYP. RC DETAIL OF 200 THK. HEAVY DUTY GRADE SLAB

SCALE 1:25



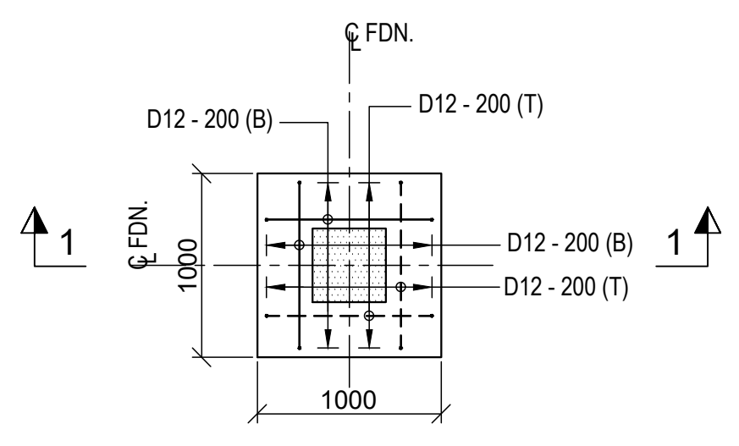
## SECTION C-C

(SHOWING RC DETAIL)
(SCALE 1:50)



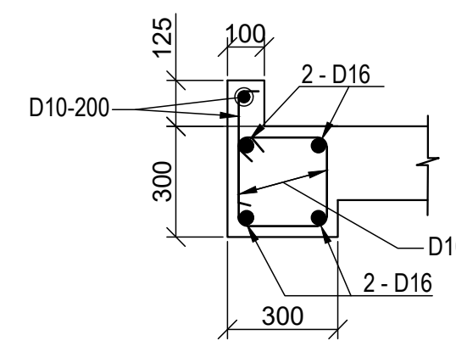
## RC DETAIL OF PIT MKD. PT1 & PT2

(SCALE 1:30)



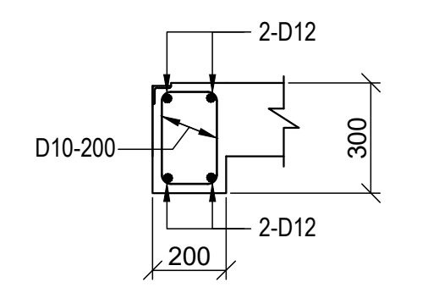
## DETAIL OF FOOTING MKD. F1

(SHOWING RC DETAIL)
(SCALE - 1:40)



## TYP. RC DETAIL OF BEAM MKD. B1

(SCALE- 1:20)



## TYP. RC DETAIL OF BEAM. MKD. B2

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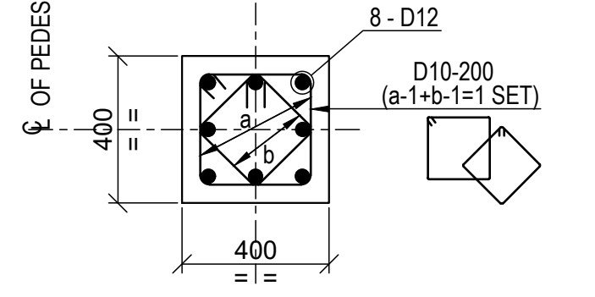
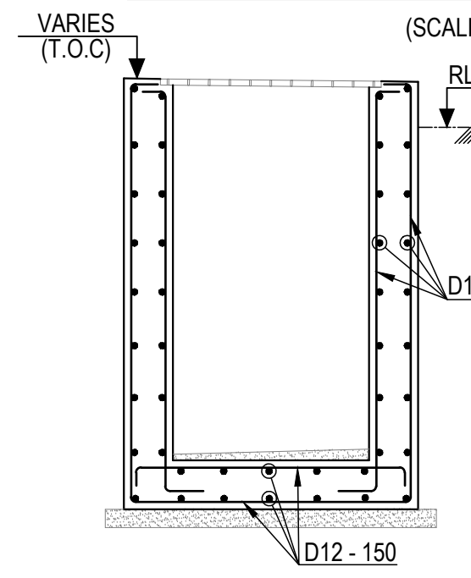


## SECTION : W - W

( RC DETAIL OF PEDESTAL MKD. P1 & P2)
(SCALE- 1:20)



## RC DETAIL OF PIT MKD. PT3

(SCALE 1:30)

LEGEND :-

R.L. - REDUCED LEVEL
T.O.C. - TOP OF CONCRETE.
F.G.L. - FINISHED GROUND LEVEL.
TYP. - TYPICAL.
P.C.C - PLAIN CEMENT CONCRETE
C - CENTRE LINE
MKD. - MARKED
B.O.F. - BOTTOM OF FOUNDATION
THK. - THICK
FDN. - FOUNDATION
DIA - DIAMETER
DWG.- DRAWING
C/C- CENTRE TO CENTRE
OS. - OILY WATER DRAINAGE PIPE
I.L - INVERT LEVEL
CE - CONDUIT EARTHING
PS - PIPE SLEEVE

NOTES :-

1. ALL DIMENSIONS ARE IN MILLIMETERS AND LEVELS AND CO-ORDINATES ARE IN METERS. UNLESS OTHERWISE STATED.
2. GRADE OF CONCRETE SHALL BE 35 MPa FOR FOUNDATION & 15 MPa FOR BLINDING CONCRETE.
3. GRADE OF REINFORCEMENT SHALL BE DEFORMED BAR 420 MPa (GRADE 60) - ASTM A615/A615M.
4. NET SAFE BEARING CAPACITY OF SOIL AT BOTTOM OF FOUNDATION CONSIDER 500KN/ SQ. M.
5. CLEAR COVER TO MAIN REINFORCEMENT SHALL BE
   a) FOUNDATION = 75 mm (BOTTOM & SIDE) 50 mm. (TOP)
   b) COLUMN & PEDESTAL = 50mm
   c) BEAM = 50mm
   d) SLAB = 20mm
6. IF ANY PART OF FOUNDATION COMES ON BACKFILL ZONE OF ADJACENT FOUNDATION OR ANY LOOSE SOIL ARE ENCOUNTERED BELOW FOUNDATION LEVEL THE SAME SHALL BE REMOVED AND FILLED WITH PCC.
7. GENERAL NOTES FOR CIVIL WORKS & STANDARD REINFORCEMENT DETAIL REFER DRAWING NO. MD-216-1000-EG-CV-DED-1001 & 1005.
8. THIS DRAWING SHALL BE READ IN CONJUNCTION WITH REFER DRAWING NO. MD-216-49PR-EG-CV-GAD-2021.
9. FOR STANDARD LADDER ARRANGEMENT REFER DRAWING NO. MD-216-1000-EG-ST-DED-1005 TO 1008.

SHEET 1 OF 2

| DOCUMENT NUMBER | REFERENCE DOCUMENTS | REV | DATE | REVISION DESCRIPTION | DRN | CHK | ENG | APP | EMP |
|---|---|---|---|---|---|---|---|---|---|
|  |  | D01 | 17/02/2023 | AS-BUILT | AMJ | SGC | SHD | PRI |  |
|  |  | C05 | 21/07/2021 | ISSUED FOR CONSTRUCTION & REVISED AS MKD. C05 | AMJ | SGC | SHD | PRI |  |
|  |  | C04 | 03/07/02021 | ISSUED FOR CONSTRUCTION | AMJ | SGC | SHD | PRI |  |
|  |  | C03 | 26/03/2021 | ISSUED FOR CONSTRUCTION | AMJ | SGC | SHD | PRI |  |
| MD-216-1000-EG-CV-DED-1003 | ANCHOR BOLTS STANDARD DETAILS. | C02 | 19/03/2021 | ISSUED FOR CONSTRUCTION & REVISED AS MKD. C02 | PDK | SGC | SHD | PRI |  |
| MD-216-1000-EG-ST-CRT-1001 | CIVIL, STRUCTURAL & ARCHITECTURAL - DESIGN CRITERIA | C01 | 04/01/2021 | ISSUED FOR CONSTRUCTION & REVISED AS MKD. C01 | PDK | SGC | SHD | PRI |  |
| MD-216-49PR-EG-ME-LAD-V553-1002 | LAYOUT OF POWER GENERATION | A01 | 30/11/2020 | ISSUED FOR REVIEW | AMJ/PDK | SGC | SHD | PRI |  |

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

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Saudi Arabian Mining Company شركة التعدين العربية السعودية

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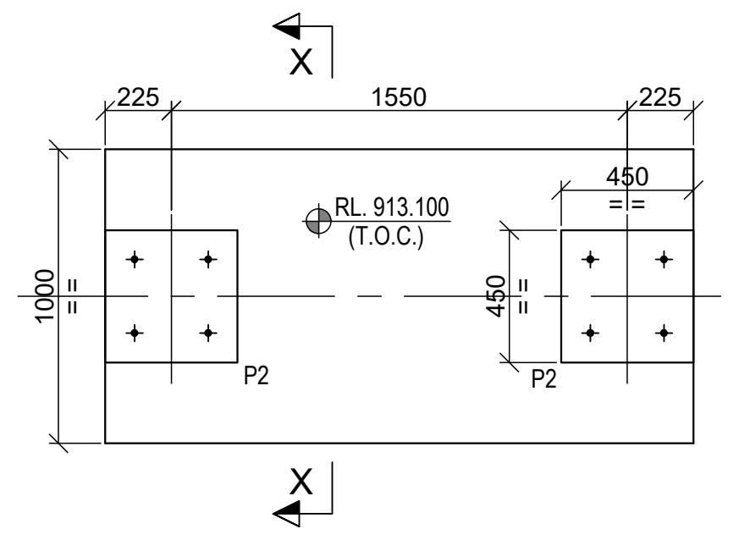
LARSEN & TOUBRO DRAWING NUMBER: 019045_008_C_G_24298_007 — REVISION 6
OUTOTEC DRAWING NUMBER: 900707 _GAU01 _WG01_MTL01 _45093 — DOCUMENT ID: OU100690016 — REVISION 6

| | | |
|---|---|---|
| DRAWN | AMJ/PDK | 10/11/2020 |
| CHECKED | SGC | 20/11/2020 |
| ENGINEER | SHD | 08/11/2020 |
| APPROVED | PRI | 30/11/2020 |
| EMPLOYER | | |
| SCALE | 1:200,100,20 | A0 |

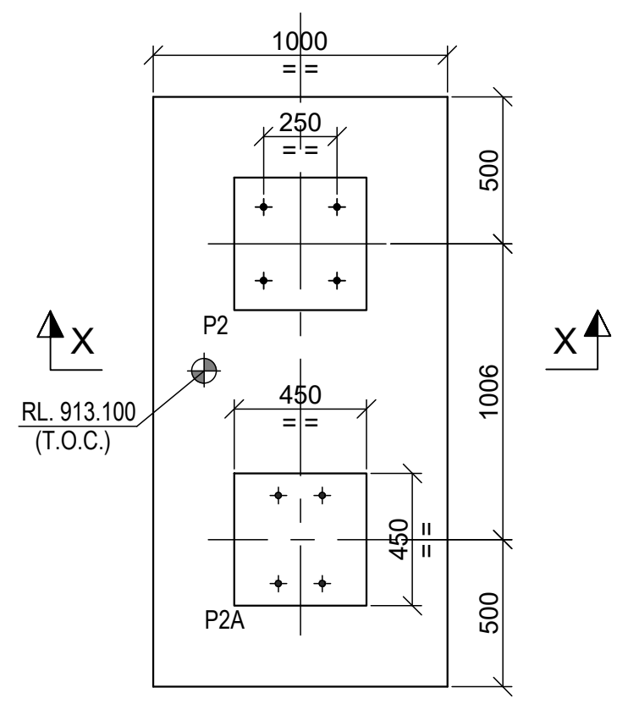
MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT
POWER PLANT MISCELLANEOUS STRUCTURE
GA AND DETAIL

DRAWING NUMBER: MD - 216 - 49PR - EG - CV - GAD - 2025 — REV D01

DRAWING NUMBER MD-216-49PR-EG-CV-GAD-2025 REV D01



DET. OF FOUNDATION MKD. F3
(SCALE 1:25)



DET. OF FOUNDATION MKD. F5
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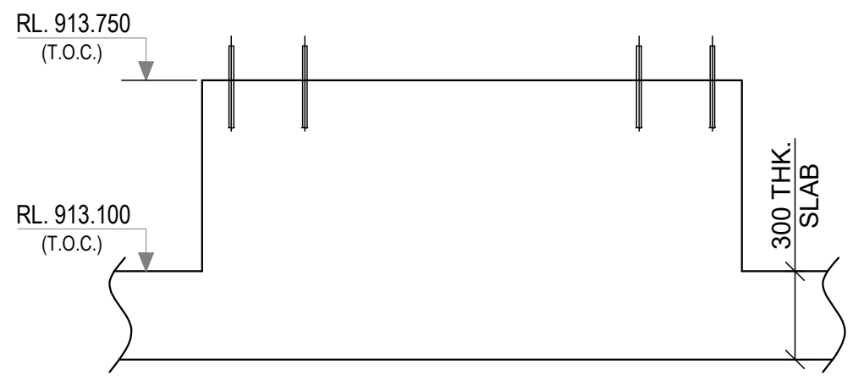

SECTION : A - A
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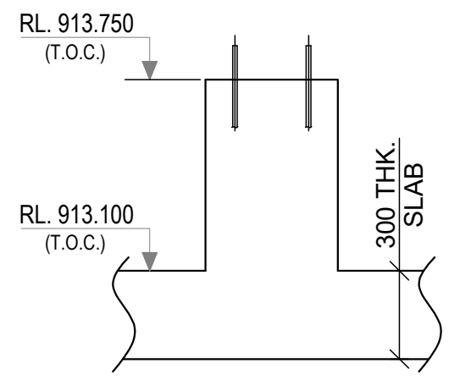

SECTION : B - B
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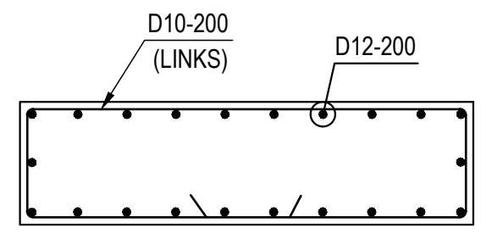
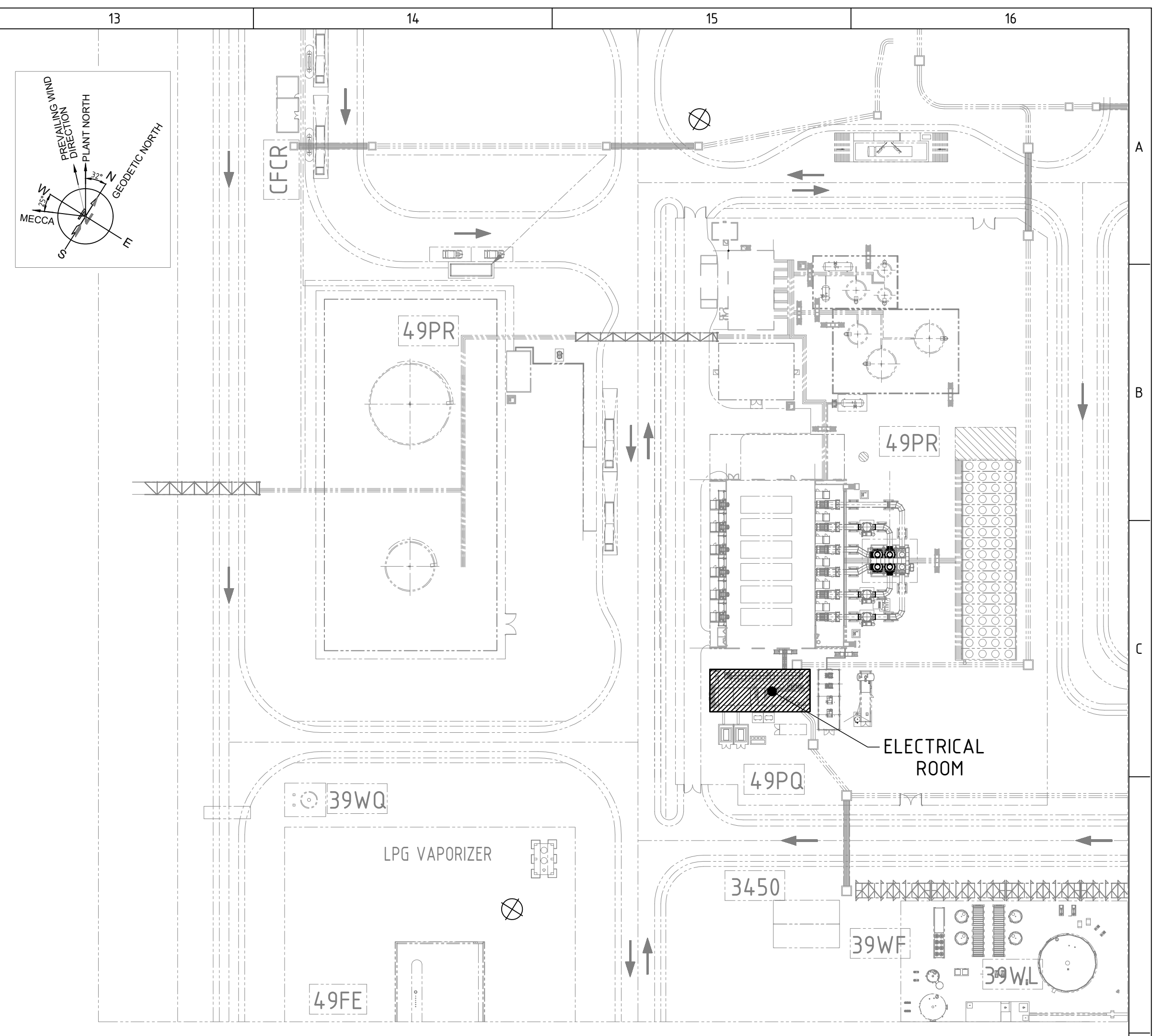
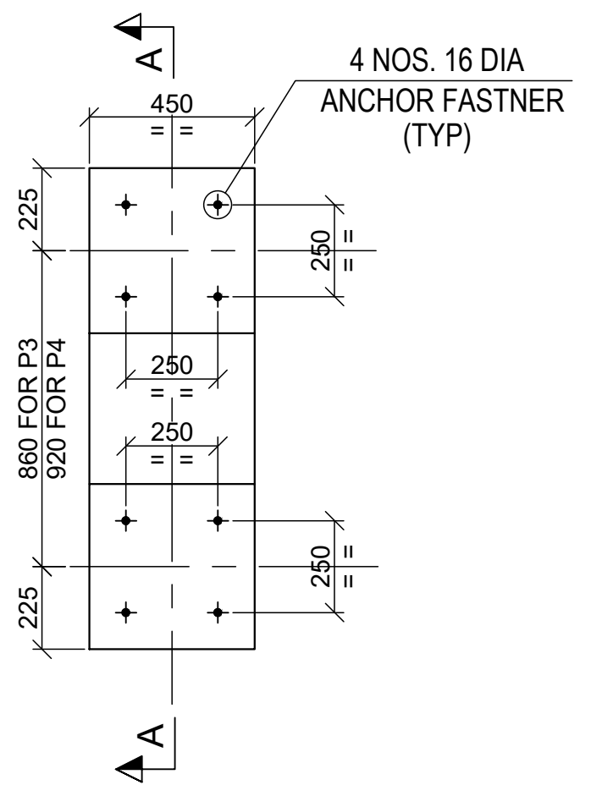


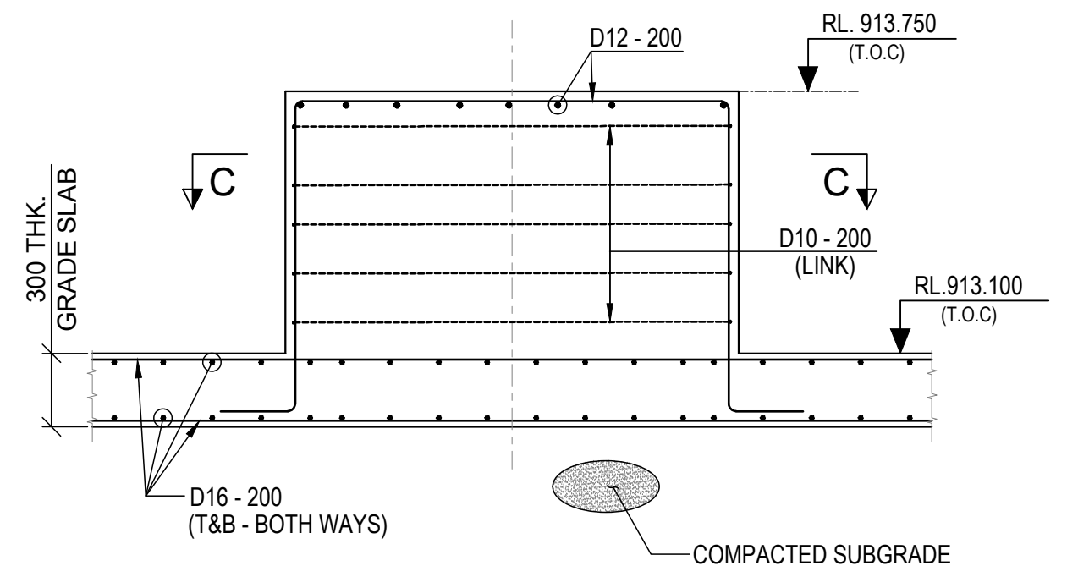
SECTION C - C
(RC DETAIL OF PEDESTAL)
(SCALE 1:30)



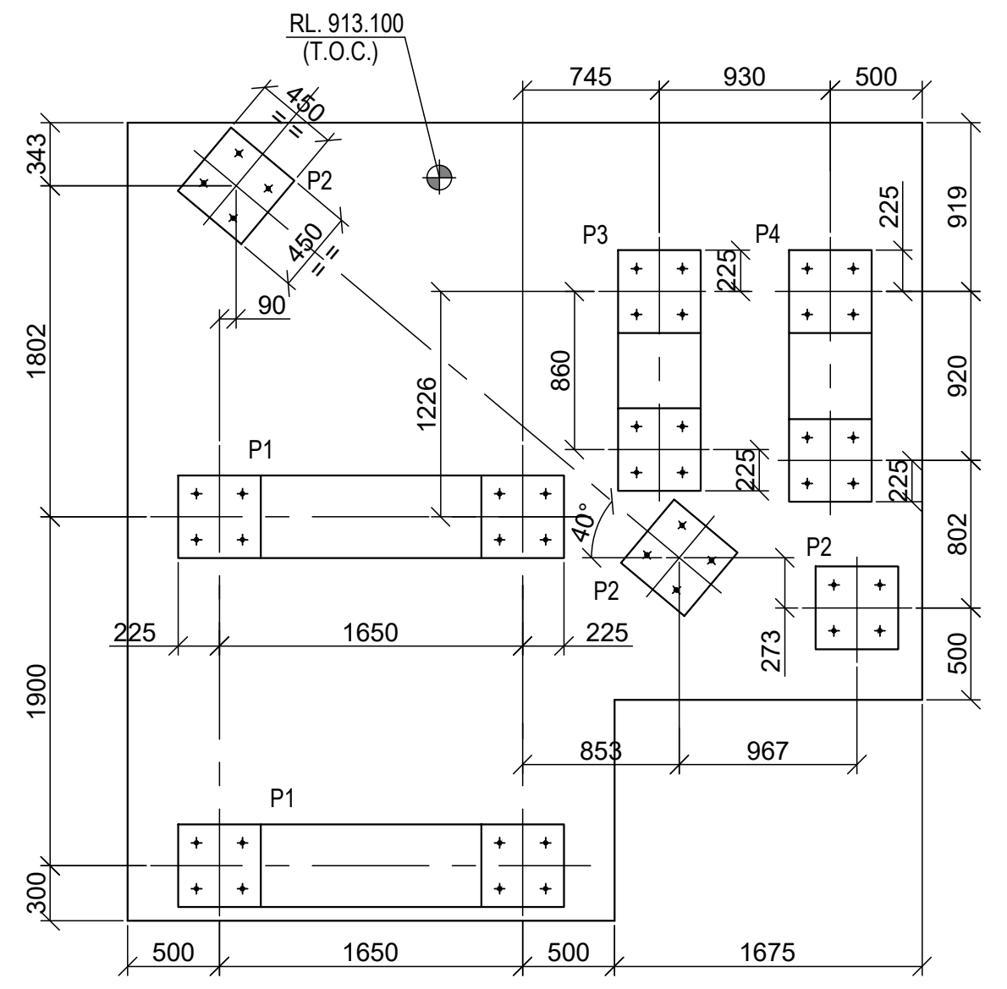
KEY PLAN
(SCALE 1:1000)



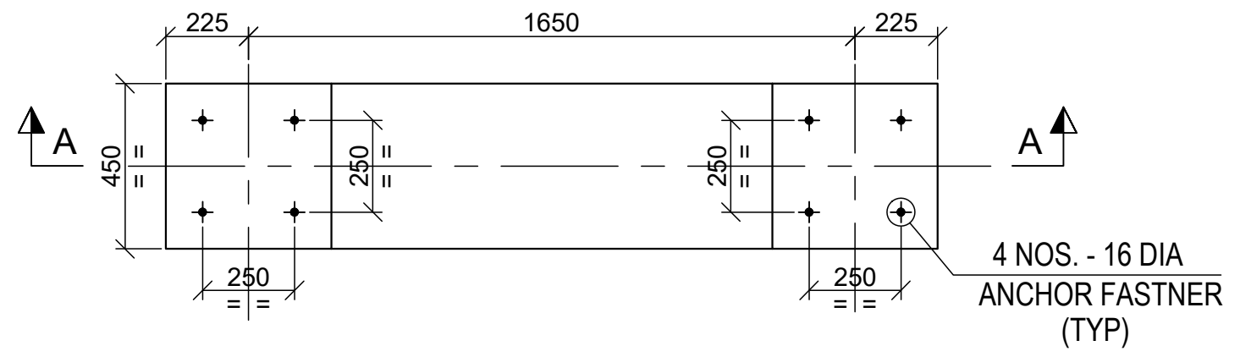
DETAIL OF PEDESTAL MKD. P3 & P4 (TYP.)
(SCALE- 1:20)



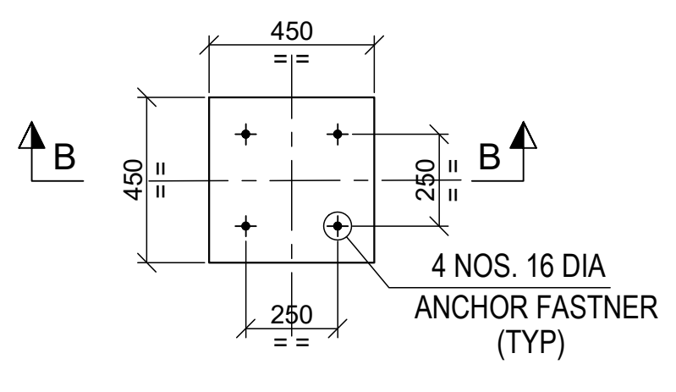
TYP. DETAIL OF PEDESTAL & FOUNDATION SLAB
(SHOWING RC DETAIL)
(SCALE 1:30)



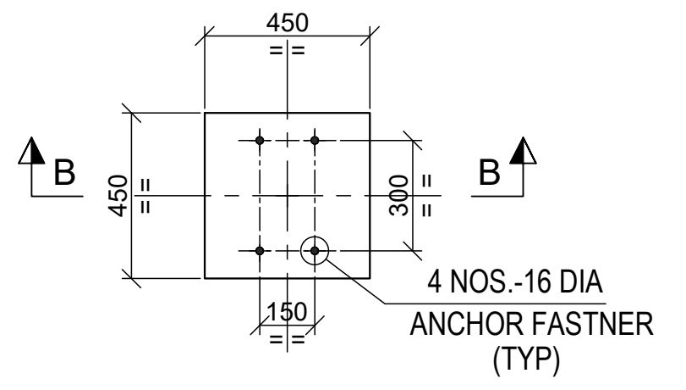
DET. OF FOUNDATION MKD. F1
(SCALE 1:40)



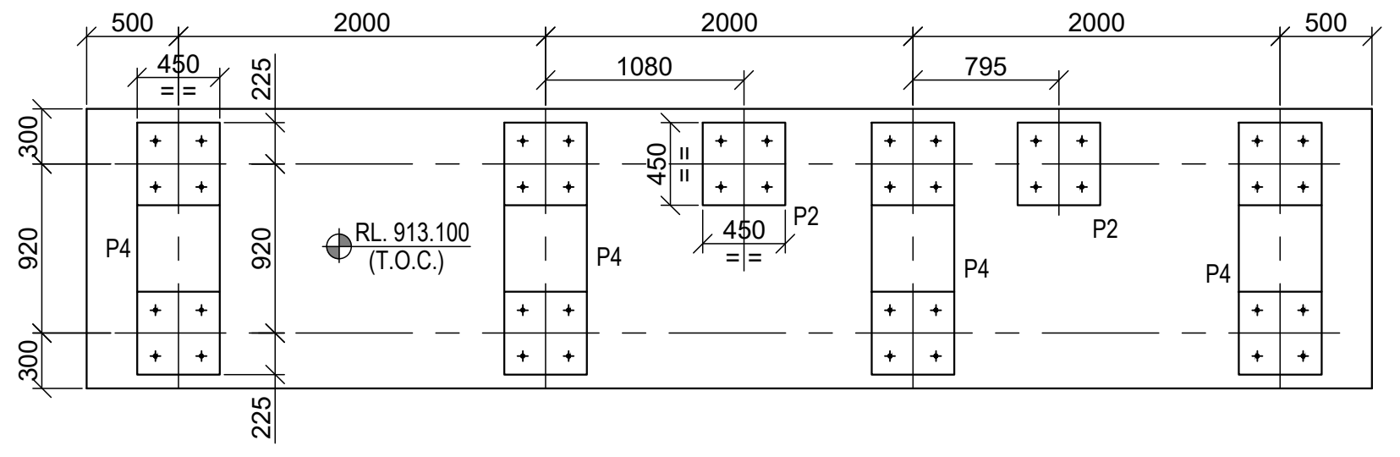
DETAIL OF PEDESTAL MKD. P1 (TYP.)
(SCALE- 1:20)



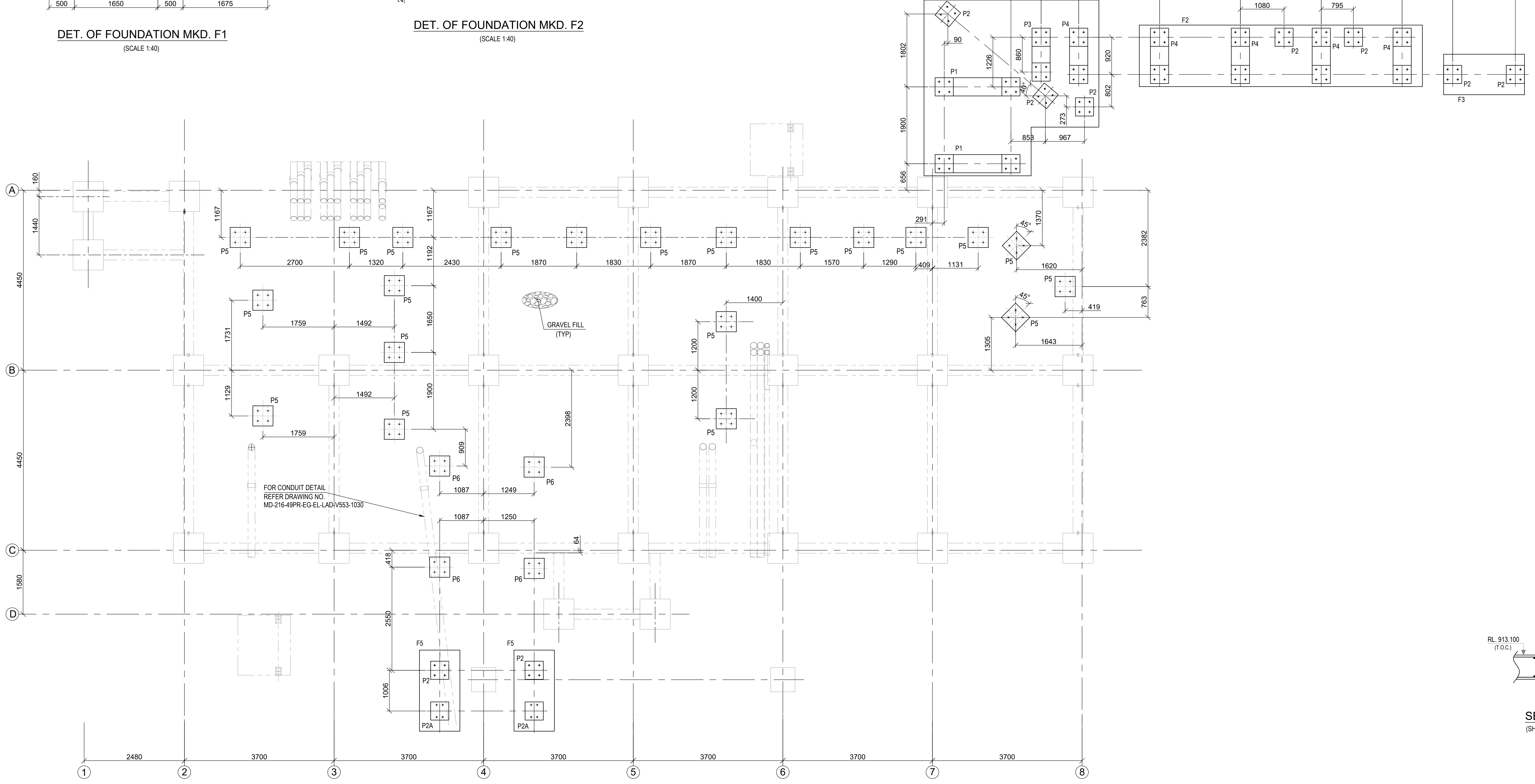
DETAIL OF PEDESTAL MKD. P2 (TYP.)
(SCALE- 1:20)



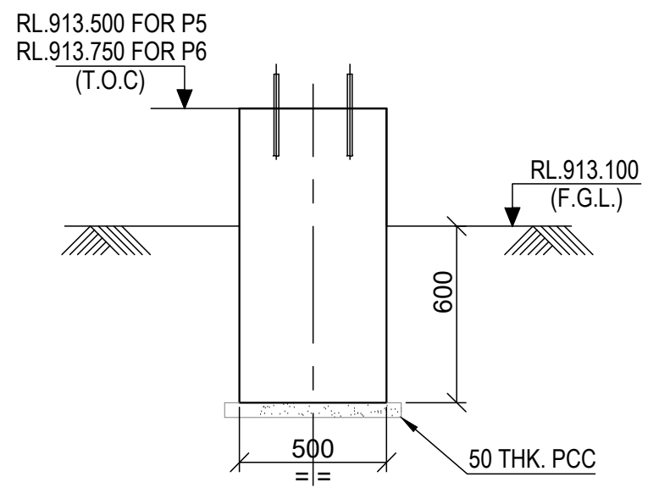
DETAIL OF PEDESTAL MKD. P2A (TYP.)
(SCALE- 1:20)



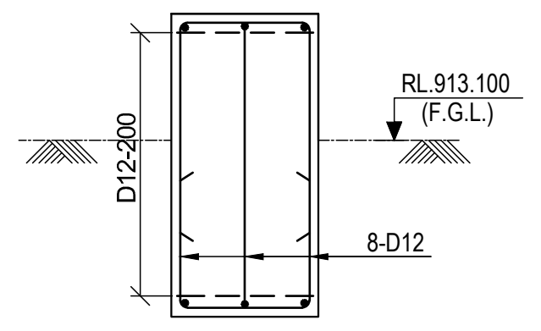
DET. OF FOUNDATION MKD. F2
(SCALE 1:40)



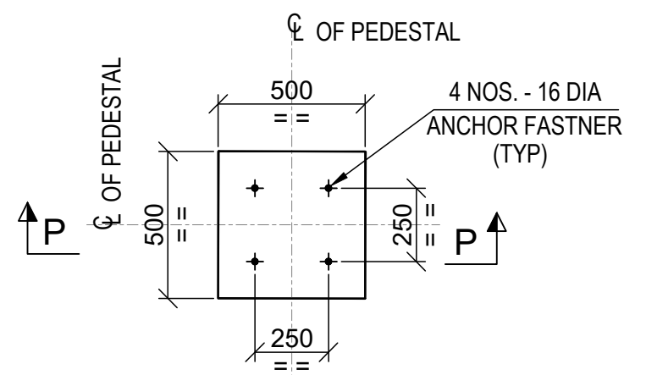
FOUNDATION LAYOUT FOR SUBSTATION
(SCALE 1:50)



SECTION P - P
(SCALE 1:25)



SECTION P1 - P1
(SCALE 1:25)



TYP. DETAIL OF PEDESTAL MKD. P5 & P6
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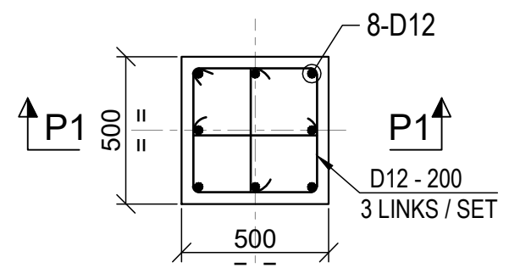
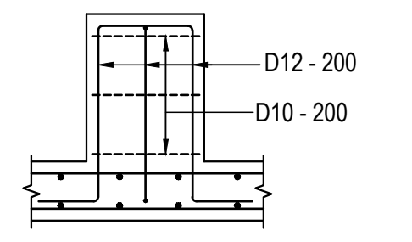
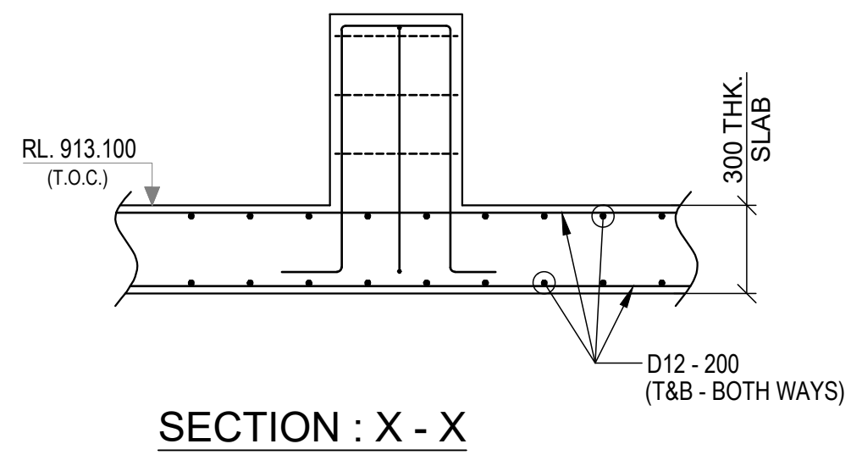


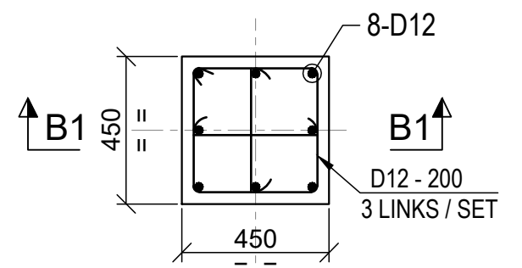
RC DETAIL OF PEDESTAL MKD. P5 & P6
(SCALE 1:25)



SECTION B1 - B1
(SCALE 1:25)



SECTION : X - X
(SHOWING RC DETAIL ONLY)
(SCALE 1:25)



RC DETAIL OF PEDESTAL MKD. P2 & P2A
(SCALE 1:25)

## LEGEND :-

R.L. - REDUCED LEVEL
T.O.C. - TOP OF CONCRETE.
F.G.L. - FINISHED GROUND LEVEL.
TYP. - TYPICAL.
P.C.C - PLAIN CEMENT CONCRETE
℄ - CENTRE LINE
MKD. - MARKED
B.O.F. - BOTTOM OF FOUNDATION
THK. - THICK
FDN. - FOUNDATION
DIA - DIAMETER
DWG.- DRAWING
C/C- CENTRE TO CENTRE
OS. - OILY WATER DRAINAGE PIPE
I.L - INVERT LEVEL
CE - CONDUIT EARTHING
PS - PIPE SLEEVE

## NOTES :-

1. ALL DIMENSIONS ARE IN MILLIMETERS AND LEVELS AND CO-ORDINATES ARE IN METERS. UNLESS OTHERWISE STATED.
2. GRADE OF CONCRETE SHALL BE 35 MPa FOR FOUNDATION & 15 MPa FOR BLINDING CONCRETE.
3. GRADE OF REINFORCEMENT SHALL BE DEFORMED BAR 420 MPa (GRADE 60) - ASTM A615/A615M.
4. NET SAFE BEARING CAPACITY OF SOIL AT BOTTOM OF FOUNDATION CONSIDER 500KN/ SQ. M.
5. CLEAR COVER TO MAIN REINFORCEMENT SHALL BE
   a) FOUNDATION = 75 mm (BOTTOM & SIDE) 50 mm. (TOP)
   b) COLUMN & PEDESTAL = 50mm
   c) BEAM = 50mm
   d) SLAB = 20mm
6. IF ANY PART OF FOUNDATION COMES ON BACKFILL ZONE OF ADJACENT FOUNDATION OR ANY LOOSE SOIL ARE ENCOUNTERED BELOW FOUNDATION LEVEL THE SAME SHALL BE REMOVED AND FILLED WITH PCC.
7. GENERAL NOTES FOR CIVIL WORKS & STANDARD REINFORCEMENT DETAIL REFER DRAWING NO. MD-216-1000-EG-CV-DED-1001 & 1005.
8. THIS DRAWING SHALL BE READ IN CONJUNCTION WITH REFER DRAWING NO. MD-216-49PR-EG-CV-GAD-2021.
9. FOR STANDARD LADDER ARRANGEMENT REFER DRAWING NO. MD-216-1000-EG-ST-DED-1005 TO 1008.

SHEET 2 OF 2

| DOCUMENT NUMBER | REFERENCE DOCUMENTS | REV | DATE | REVISION DESCRIPTION | DRN | CHK | ENG | APP | EMP |
|---|---|---|---|---|---|---|---|---|---|
| | | D01 | 17/02/2023 | AS-BUILT | AMJ | SGC | SHD | PRI | |
| | | C05 | 21/07/2021 | ISSUED FOR CONSTRUCTION | AMJ | SGC | SHD | PRI | |
| | | C04 | 03/07/2021 | ISSUED FOR CONSTRUCTION & REVISED AS MKD. C04 | AMJ | SGC | SHD | PRI | |
| | | C03 | 26/03/2021 | ISSUED FOR CONSTRUCTION & REVISED AS MKD. C03 | AMJ/PDK | SGC | SHD | PRI | |
| MD-216-1000-EG-CV-DED-1003 | ANCHOR BOLTS STANDARD DETAILS. | C02 | 19/03/2021 | ISSUED FOR CONSTRUCTION | AMJ/PDK | SGC | SHD | PRI | |
| MD-216-1000-EG-ST-CRT-1001 | CIVIL, STRUCTURAL & ARCHITECTURAL - DESIGN CRITERIA | C01 | 04/01/2021 | ISSUED FOR CONSTRUCTION & REVISED AS MKD. | PDK | SGC | SHD | PRI | |
| MD-216-49PR-EG-ME-LAD-V553-1002 | LAYOUT OF POWER GENERATION | A01 | 30/11/2020 | ISSUED FOR REVIEW | AMJ/PDK | SGC | SHD | PRI | |

| DOCUMENT NUMBER | REFERENCE DOCUMENTS | REV | DATE | REVISION DESCRIPTION | DRN | CHK | ENG | APP | EMP |
|---|---|---|---|---|---|---|---|---|---|

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

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Outotec

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LARSEN & TOUBRO DRAWING NUMBER: 019045_008_C_G_24298_007 — REVISION 6
OUTOTEC DRAWING NUMBER: 900707 _ GAU01 _ WG01 _ MTL01 _ 45093 — DOCUMENT ID: OU100690016 — REVISION 6

| | | |
|---|---|---|
| DRAWN | AMJ/PDK | 10/11/2020 |
| CHECKED | SGC | 20/11/2020 |
| ENGINEER | SHD | 08/11/2020 |
| APPROVED | PRI | 30/11/2020 |
| EMPLOYER | | |
| SCALE | 1:200,100,20 | A0 |

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT
POWER PLANT MISCELLANEOUS STRUCTURE
GA AND DETAIL

DRAWING NUMBER: MD - 216 - 49PR - EG - CV - GAD - 2025 — REV D01